Conditional Probability & Bayes' Rule

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

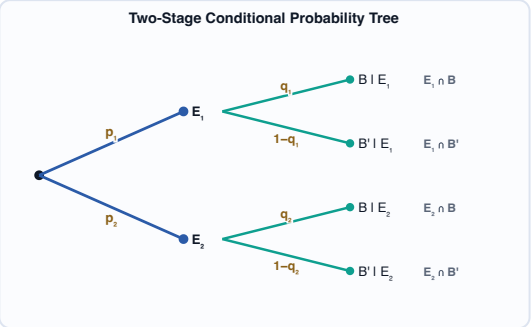
QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 A	Q2 A	Q3 A	Q4 A	Q5 A
Q6 A	Q7 A	Q8 A	Q9 D	Q10 D
Q11 B	Q12 B	Q13 C	Q14 A	Q15 B
Q16 D	Q17 D	Q18 A	Q19 A	Q20 B
Q21 D	Q22 D	Q23 C	Q24 D	Q25 D

Conditional Probability & Bayes' Rule Practice Exercises

1 MCQ

In a dataset, the prior probability of class C_1 is $P(C_1) = 0.10$. What is the probability of the complement class $P(\neg C_1)$?



A 0.90 ✓

B 0.95

C 0.05

D 1.10

WORKED SOLUTION & VERIFICATION

By the complement rule of probability: $P(\neg C_1) = 1 - P(C_1) = 1 - 0.10 = 0.90$.

2 MCQ

In a dataset, the prior probability of class C_1 is $P(C_1) = 0.20$. What is the probability of the complement class $P(\neg C_1)$?

A 0.80 ✓

B 0.90

C 0.10

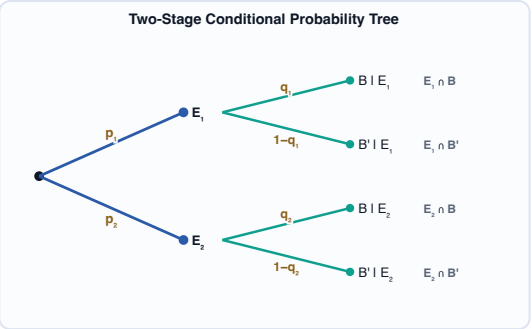
D 1.20

WORKED SOLUTION & VERIFICATION

By the complement rule of probability: $P(\neg C_1) = 1 - P(C_1) = 1 - 0.20 = 0.80$.

3 MCQ

In a dataset, the prior probability of class C_1 is $P(C_1) = 0.60$. What is the probability of the complement class $P(\neg C_1)$?



- **A** 0.40 ✓
- **B** 0.70
- **C** 0.30
- **D** 1.60

WORKED SOLUTION & VERIFICATION

By the complement rule of probability: $P(\neg C_1) = 1 - P(C_1) = 1 - 0.60 = 0.40$.

4 MCQ

A machine learning diagnostic test has sensitivity $P(\text{Positive}|\text{Defect}) = 0.90$. If the defect rate in production is $P(\text{Defect}) = 0.05$, find the joint probability of a defect and a positive test result $P(\text{Defect} \cap \text{Positive})$.

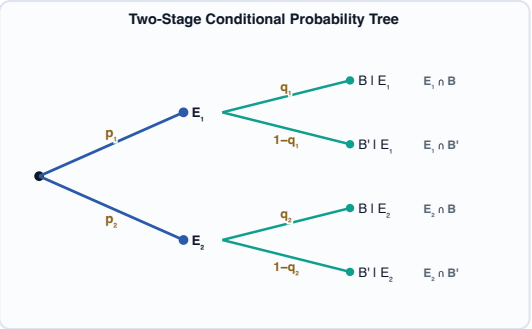
- **A** 0.0450 ✓
- **B** 0.9500
- **C** 0.8500
- **D** 0.0050

WORKED SOLUTION & VERIFICATION

By the multiplication rule of conditional probability: $P(\text{Defect} \cap \text{Positive}) = P(\text{Defect}) \times P(\text{Positive}|\text{Defect}) = 0.05 \times 0.90 = 0.0450$.

5 MCQ

A machine learning diagnostic test has sensitivity $P(\text{Positive}|\text{Defect}) = 0.90$. If the defect rate in production is $P(\text{Defect}) = 0.10$, find the joint probability of a defect and a positive test result $P(\text{Defect} \cap \text{Positive})$.



- **A** 0.0900 ✓
- **B** 1.0000
- **C** 0.8000
- **D** 0.0100

WORKED SOLUTION & VERIFICATION

By the multiplication rule of conditional probability: $P(\text{Defect} \cap \text{Positive}) = P(\text{Defect}) \times P(\text{Positive}|\text{Defect}) = 0.10 \times 0.90 = 0.0900$.

6 MCQ

A machine learning diagnostic test has sensitivity $P(\text{Positive}|\text{Defect}) = 0.90$. If the defect rate in production is $P(\text{Defect}) = 0.15$, find the joint probability of a defect and a positive test result $P(\text{Defect} \cap \text{Positive})$.

- **A** 0.1350 ✓
- **B** 1.0500
- **C** 0.7500
- **D** 0.0150

WORKED SOLUTION & VERIFICATION

By the multiplication rule of conditional probability: $P(\text{Defect} \cap \text{Positive}) = P(\text{Defect}) \times P(\text{Positive}|\text{Defect}) = 0.15 \times 0.90 = 0.1350$.

7 MCQ

A machine learning diagnostic test has sensitivity $P(\text{Positive}|\text{Defect}) = 0.90$. If the defect rate in production is $P(\text{Defect}) = 0.20$, find the joint probability of a defect and a positive test result $P(\text{Defect} \cap \text{Positive})$.

- **A** 0.1800 ✓
- **B** 1.1000
- **C** 0.7000
- **D** 0.0200

WORKED SOLUTION & VERIFICATION

By the multiplication rule of conditional probability: $P(\text{Defect} \cap \text{Positive}) = P(\text{Defect}) \times P(\text{Positive}|\text{Defect}) = 0.20 \times 0.90 = 0.1800$.

8 MCQ

A machine learning diagnostic test has sensitivity $P(\text{Positive}|\text{Defect}) = 0.90$. If the defect rate in production is $P(\text{Defect}) = 0.25$, find the joint probability of a defect and a positive test result $P(\text{Defect} \cap \text{Positive})$.

- **A** 0.2250 ✓
- **B** 1.1500
- **C** 0.6500
- **D** 0.0250

WORKED SOLUTION & VERIFICATION

By the multiplication rule of conditional probability: $P(\text{Defect} \cap \text{Positive}) = P(\text{Defect}) \times P(\text{Positive}|\text{Defect}) = 0.25 \times 0.90 = 0.2250$.

9 MCQ

An Ordinary Least Squares linear model $\hat{p}(t) = 2t + 11$ is evaluated at $t = 3$. Determine the predicted response $\hat{p}(3)$.

- **A** 19
- **B** 21
- **C** 6
- **D** **17** ✓

WORKED SOLUTION & VERIFICATION

Evaluating $2(3) + 11 = 17$.

10 MCQ

Considering a dataset, the prior probability of class C_1 is $P(C_1) = 0.20$. What is the probability of the complement class $P(\neg C_1)$?

- **A** 0.90
- **B** 0.10
- **C** 1.20
- **D** 0.80 ✓

WORKED SOLUTION & VERIFICATION

By the complement rule of probability: $P(\neg C_1) = 1 - P(C_1) = 1 - 0.20 = 0.80$.

11 MCQ

In gradient descent on loss $h(u)$, parameter $u_0 = 44$ is updated with learning rate $\eta = 0.1$ and gradient $\nabla h(u_0) = 70$. Determine the updated parameter $u_1 = u_0 - \eta \nabla h(u_0)$.

- **A** 61
- **B** **37** ✓
- **C** 30
- **D** 41

WORKED SOLUTION & VERIFICATION

Computing $u_1 = 44 - 0.1(70) = 37$.

12 MCQ

Examining a dataset, the prior probability of class C_1 is $P(C_1) = 0.60$. What is the probability of the complement class $P(\neg C_1)$?

A 0.70	B 0.40 ✓
C 0.30	D 1.60

WORKED SOLUTION & VERIFICATION

By the complement rule of probability: $P(\neg C_1) = 1 - P(C_1) = 1 - 0.60 = 0.40$.

13 MCQ

A medical test for a rare disease has a 99\% accuracy rate in detecting the disease when it is present (True Positive Rate) and a 95\% accuracy rate in identifying individuals without the disease (True Negative Rate). If the prevalence of the disease in the population is 0.5\%, what is the probability that a person who tests positive actually has the disease?

A Approximately 8.7\%	B Approximately 1.0\%
C **Approximately 9.1\%** ✓	D Approximately 0.5\%

WORKED SOLUTION & VERIFICATION

Let D be the event that a person has the disease, and P be the event that the person tests positive.
We are given:
P(P|D) = 0.99 (True Positive Rate)
P(not P|not D) = 0.95 (True Negative Rate), which means P(P|not D) = 1 - 0.95 = 0.05 (False Positive Rate)
P(D) = 0.005 (Prevalence of the disease)
From P(D), we know P(not D) = 1 - P(D) = 1 - 0.005 = 0.995.
We want to find P(D|P), the probability that a person has the disease given a positive test result.
Using Bayes' Theorem:
P(D|P) = [P(P|D) * P(D)] / P(P)
To find P(P), we use the law of total probability:
P(P) = P(P|D) * P(D) + P(P|not D) * P(not D)
P(P) = (0.99 * 0.005) + (0.05 * 0.995)
P(P) = 0.00495 + 0.04975
P(P) = 0.0547
Now, substitute this back into Bayes' Theorem:
P(D|P) = (0.99 * 0.005) / 0.0547
P(D|P) = 0.00495 / 0.0547
P(D|P) \approx 0.09049
Converting to a percentage, this is approximately 9.05\%, which rounds to 9.1\%.

14 MCQ

A medical diagnostic test for a rare disease is designed to be highly sensitive and specific. The disease affects 1 in every 1000 individuals in the population. The test has a sensitivity of 99% (meaning it correctly identifies 99% of individuals who have the disease) and a specificity of 95% (meaning it correctly identifies 95% of individuals who do not have the disease). If a randomly selected individual from the population tests positive, what is the probability that they actually have the disease?

- A **Approximately** 0.019 ✓
- B Approximately 0.99
- C Approximately 0.05
- D Approximately 0.95

WORKED SOLUTION & VERIFICATION

Let D be the event that an individual has the disease, and P be the event that the individual tests positive.
We are given the following probabilities:
- Prior probability of having the disease: $P(D) = \frac{1}{1000} = 0.001$
- Prior probability of not having the disease: $P(D^c) = 1 - P(D) = 1 - 0.001 = 0.999$
- Sensitivity (true positive rate): $P(P|D) = 0.99$
- Specificity (true negative rate): $P(P^c|D^c) = 0.95$

From specificity, we can derive the false positive rate: $P(P|D^c) = 1 - P(P^c|D^c) = 1 - 0.95 = 0.05$.
We want to find the probability that an individual actually has the disease given a positive test result, which is $P(D|P)$. We can use Bayes' Theorem:

$$P(D|P) = \frac{P(P|D)P(D)}{P(P)}$$

To find $P(P)$, the overall probability of testing positive, we can use the law of total probability:

$$P(P) = P(P|D)P(D) + P(P|D^c)P(D^c)$$

Substitute the given values:

$$P(P) = (0.99)(0.001) + (0.05)(0.999)$$

$$P(P) = 0.00099 + 0.04995$$

$$P(P) = 0.05094$$

Now, substitute $P(P)$ back into Bayes' Theorem:

$$P(D|P) = \frac{(0.99)(0.001)}{0.05094}$$

$$P(D|P) = \frac{0.00099}{0.05094}$$

$$P(D|P) \approx 0.0194346$$

Therefore, the probability that a person who tests positive actually has the disease is approximately 0.0194, which is closest to 0.019.

15 MCQ

In an applied problem, a machine learning diagnostic test has sensitivity $P(\text{Positive}|\text{Defect}) = 0.90$. If the defect rate in production is $P(\text{Defect}) = 0.15$, find the joint probability of a defect and a positive test result $P(\text{Defect} \cap \text{Positive})$.

A 1.0500

B 0.1350 ✓

C 0.7500

D 0.0150

WORKED SOLUTION & VERIFICATION

By the multiplication rule of conditional probability: $P(\text{Defect} \cap \text{Positive}) = P(\text{Defect}) \times P(\text{Positive}|\text{Defect}) = 0.15 \times 0.90 = 0.1350$.

16 MCQ

Suppose a machine learning diagnostic test has sensitivity $P(\text{Positive}|\text{Defect}) = 0.90$. If the defect rate in production is $P(\text{Defect}) = 0.20$, find the joint probability of a defect and a positive test result $P(\text{Defect} \cap \text{Positive})$.

A 1.1000

B 0.7000

C 0.0200

D 0.1800 ✓

WORKED SOLUTION & VERIFICATION

By the multiplication rule of conditional probability: $P(\text{Defect} \cap \text{Positive}) = P(\text{Defect}) \times P(\text{Positive}|\text{Defect}) = 0.20 \times 0.90 = 0.1800$.

17 MCQ

An observed machine learning diagnostic test has sensitivity $P(\text{Positive}|\text{Defect}) = 0.90$. If the defect rate in production is $P(\text{Defect}) = 0.25$, find the joint probability of a defect and a positive test result $P(\text{Defect} \cap \text{Positive})$.

A 0.0250

B 1.1500

C 0.6500

D 0.2250 ✓

WORKED SOLUTION & VERIFICATION

By the multiplication rule of conditional probability: $P(\text{Defect} \cap \text{Positive}) = P(\text{Defect}) \times P(\text{Positive}|\text{Defect}) = 0.25 \times 0.90 = 0.2250$.

18 MCQ

Consider a machine learning diagnostic test has sensitivity $P(\text{Positive}|\text{Defect}) = 0.90$. If the defect rate in production is $P(\text{Defect}) = 0.30$, find the joint probability of a defect and a positive test result $P(\text{Defect} \cap \text{Positive})$.

- A 0.2700 ✓
- B 1.2000
- C 0.6000
- D 0.0300

WORKED SOLUTION & VERIFICATION

By the multiplication rule of conditional probability: $P(\text{Defect} \cap \text{Positive}) = P(\text{Defect}) \times P(\text{Positive}|\text{Defect}) = 0.30 \times 0.90 = 0.2700$.

19 MCQ

In anomaly detection, anomaly prior $P(A) = 0.02$, detection sensitivity $P(+|A) = 0.95$, and false alarm rate $P(+|\neg A) = 0.05$. By Bayes' theorem, what is the posterior probability $P(A|+)$ that an alarm is a true anomaly?

- A 0.279 ✓
- B 0.140
- C 0.950
- D 0.391

WORKED SOLUTION & VERIFICATION

By Bayes' Rule: $P(A|+) = \frac{P(+|A)P(A)}{P(+|A)P(A)+P(+|\neg A)P(\neg A)} = \frac{0.95\times0.02}{0.95(0.02)+0.05(0.98)} = \frac{0.0190}{0.0680} = 0.279$.

20 MCQ

Considering anomaly detection, anomaly prior $P(A) = 0.04$, detection sensitivity $P(+|A) = 0.95$, and false alarm rate $P(+|\neg A) = 0.05$. By Bayes' theorem, what is the posterior probability $P(A|+)$ that an alarm is a true anomaly?

- A 0.221
- B 0.442 ✓
- C 0.950
- D 0.619

WORKED SOLUTION & VERIFICATION

By Bayes' Rule: $P(A|+) = \frac{P(+|A)P(A)}{P(+|A)P(A)+P(+|\neg A)P(\neg A)} = \frac{0.95\times0.04}{0.95(0.04)+0.05(0.96)} = \frac{0.0380}{0.0860} = 0.442$.

21 MCQ

Examining anomaly detection, anomaly prior $P(A) = 0.06$, detection sensitivity $P(+|A) = 0.95$, and false alarm rate $P(+|\neg A) = 0.05$. By Bayes' theorem, what is the posterior probability $P(A|+)$ that an alarm is a true anomaly?

- A 0.950
- B 0.767
- C 0.274
- D 0.548 ✓

WORKED SOLUTION & VERIFICATION

By Bayes' Rule: $P(A|+) = \frac{P(+|A)P(A)}{P(+|A)P(A)+P(+|\neg A)P(\neg A)} = \frac{0.95\times0.06}{0.95(0.06)+0.05(0.94)} = \frac{0.0570}{0.1040} = 0.548$.

22 MCQ

For the configuration in anomaly detection, anomaly prior $P(A) = 0.08$, detection sensitivity $P(+|A) = 0.95$, and false alarm rate $P(+|\neg A) = 0.05$. By Bayes' theorem, what is the posterior probability $P(A|+)$ that an alarm is a true anomaly?

A 0.872

B 0.311

C 0.950

D 0.623 ✓

WORKED SOLUTION & VERIFICATION

By Bayes' Rule: $P(A|+) = \frac{P(+|A)P(A)}{P(+|A)P(A)+P(+|\neg A)P(\neg A)} = \frac{0.95\times0.08}{0.95(0.08)+0.05(0.92)} = \frac{0.0760}{0.1220} = 0.623.$

23 MCQ

Analyzing anomaly detection, anomaly prior $P(A) = 0.10$, detection sensitivity $P(+|A) = 0.95$, and false alarm rate $P(+|\neg A) = 0.05$. By Bayes' theorem, what is the posterior probability $P(A|+)$ that an alarm is a true anomaly?

A 0.340

B 0.950

C 0.679 ✓

D 0.951

WORKED SOLUTION & VERIFICATION

By Bayes' Rule: $P(A|+) = \frac{P(+|A)P(A)}{P(+|A)P(A)+P(+|\neg A)P(\neg A)} = \frac{0.95\times0.10}{0.95(0.10)+0.05(0.90)} = \frac{0.0950}{0.1400} = 0.679.$

24 MCQ

Within anomaly detection, anomaly prior $P(A) = 0.12$, detection sensitivity $P(+|A) = 0.95$, and false alarm rate $P(+|\neg A) = 0.05$. By Bayes' theorem, what is the posterior probability $P(A|+)$ that an alarm is a true anomaly?

A 0.361

B 0.950

C 1.011

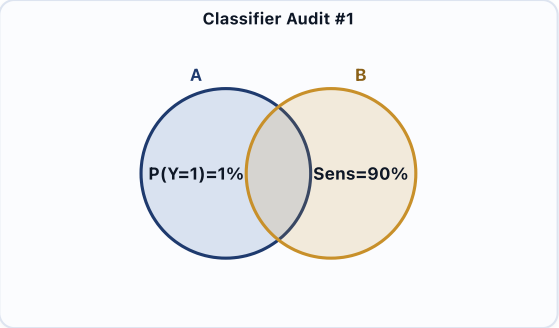
D 0.722 ✓

WORKED SOLUTION & VERIFICATION

By Bayes' Rule: $P(A|+) = \frac{P(+|A)P(A)}{P(+|A)P(A)+P(+|\neg A)P(\neg A)} = \frac{0.95\times0.12}{0.95(0.12)+0.05(0.88)} = \frac{0.1140}{0.1580} = 0.722.$

25 MCQ

In classifier audit #1, positive-class prevalence is $P(Y = 1) = 1\%$, sensitivity is $P(\hat{Y} = 1 \mid Y = 1) = 90\%$, and false-positive rate is $P(\hat{Y} = 1 \mid Y = 0) = 5\%$. By Bayes' Rule, find the precision $P(Y = 1 \mid \hat{Y} = 1)$.



A $\frac{9}{10}$

B $\frac{9}{1000}$

C $\frac{11}{13}$

D $\frac{2}{13}$ ✓

WORKED SOLUTION & VERIFICATION

By Bayes' Rule, $P(Y = 1 \mid \hat{Y} = 1) = \frac{(1)(90)}{(1)(90)+(99)(5)} = \frac{90}{585} = \frac{2}{13}$.